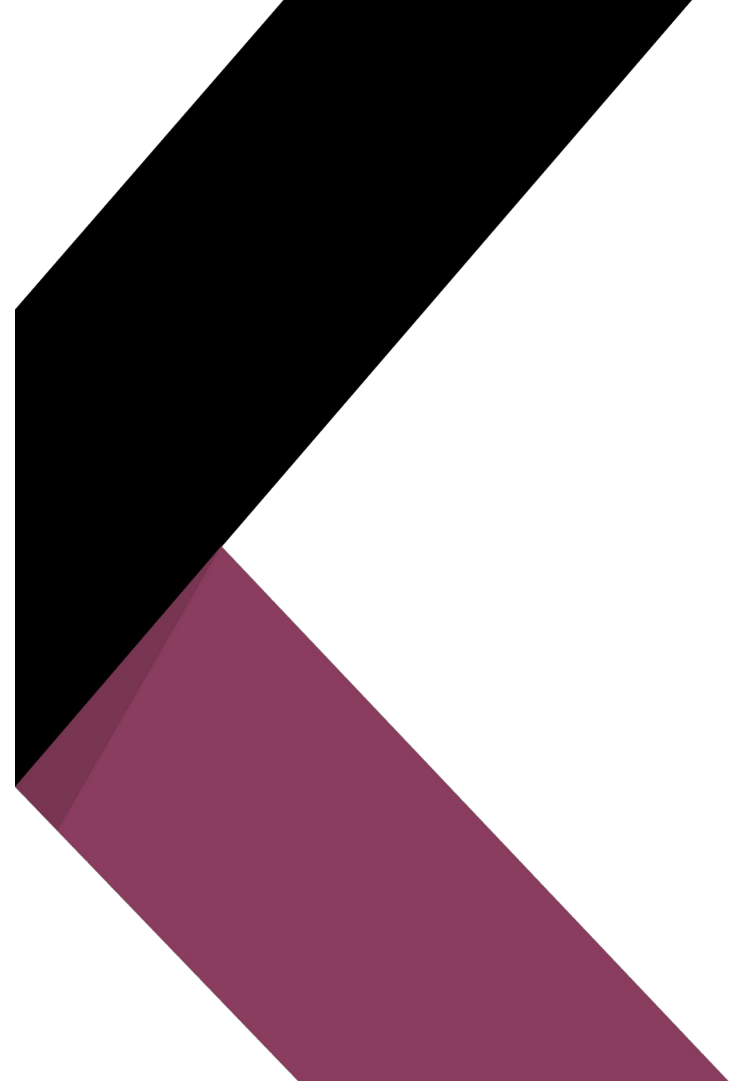


Software analysis

INCA - MDA





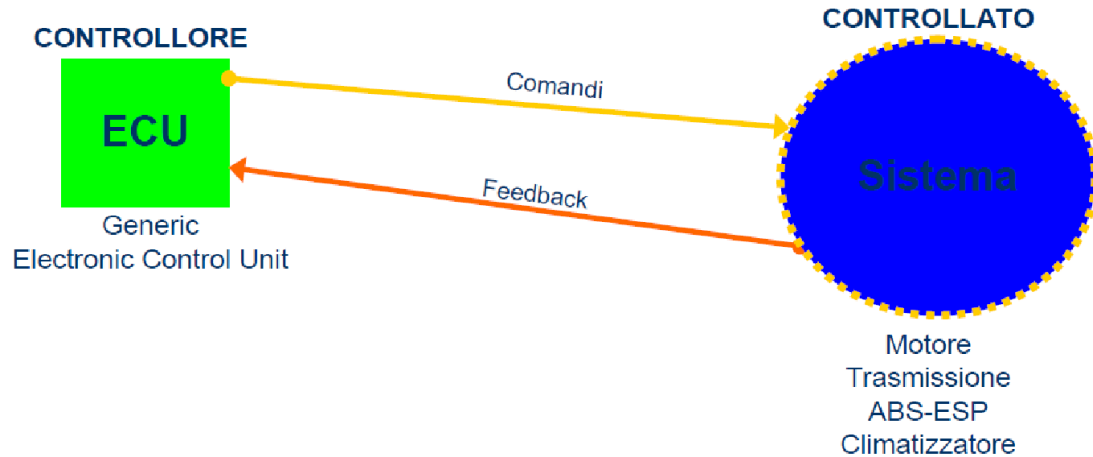
Introduction



Vehicle's

Control

In modern vehicles the systems controlled by a dedicated ECU that interacts with those of other systems





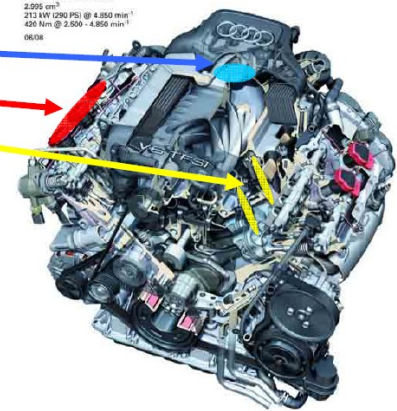
ECM (Engine Control Module)

The ECM is the ECU dedicated to engine control and its main tasks are :

- Combustion control
- Management of the requested power
- Emission control

Attuatori

- Corpo Farfallato
- Candele
- Iniettori
- ...





Software and

Calibration

- **SOFTWARE:** the software of the ECM is composed by a firmware and an application; the application contains all the algorithms and the engine control strategies. It is normally supplied with default settings (calibrations)
- **CALIBRATION:** the calibration procedure is the operation through which the software is customized for different applications (this makes it possible to use the same software for different applications)



Development ECU Vs

Production ECU

There are two different types of ECUs :

- Opened or development
- Closed or production

The main difference between the two types is in the number of installed memories



Production ECU

The production unit has the following features:

- ECU flash: processor memory that is programmed with the complete sw
- ECU Ram: supporting memory
- EEPROM: part of the memory on which the control unit saves data before sw



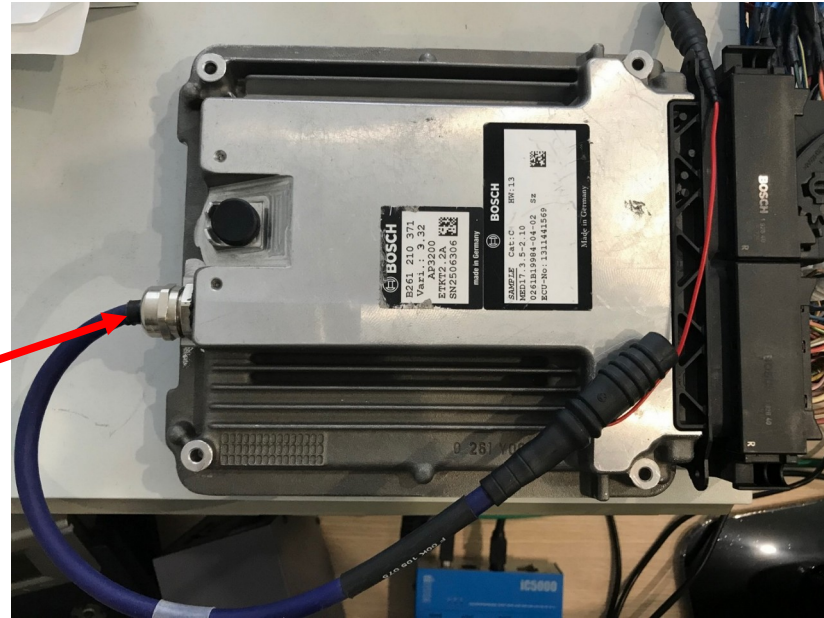


Development ECU

The development unit has the following features :

- ECU flash and EEPROM are the same of the production one
- ETK flash: additional memory that stores a page with calibrations during ECU development
- ETK Ram: volatile memory that contains two calibration pages "Working page" and "Reference page" with the possibility to switch from one to the other on the fly

ETK cable connected to the
ETK Ram memory





INCA

INtegrated Calibration and Measurement System

INCA

INCA is a diagnostic tool that allows you to :

- Read the variables of the software present on the ECM (Engine Control Module)
- Change the calibration parameters
- Acquire the results of a measurement/calibration
- Download the sw on the ECU memory



ETAS Hardware

INCA uses some specific interfaces produced by ETAS to communicate with the ECUs; below there are the images of two ETAS hardware :

ES 592



ES 581





ETAS - ES592

The ES592 interface is a more complete version of the ETAS peripheral as it allows connection with the development ECU via an ETK cable or via a CAN network (CCP connection) with CAN production control units





ETAS - ES581

The ES581 interface represents a simpler version of the ETAS peripherals and provides connection to the ECU (both production and development) only via CAN network (CCP connection).

With this hardware it is also possible to monitor the CAN Network by the CAN monitoring functionality.

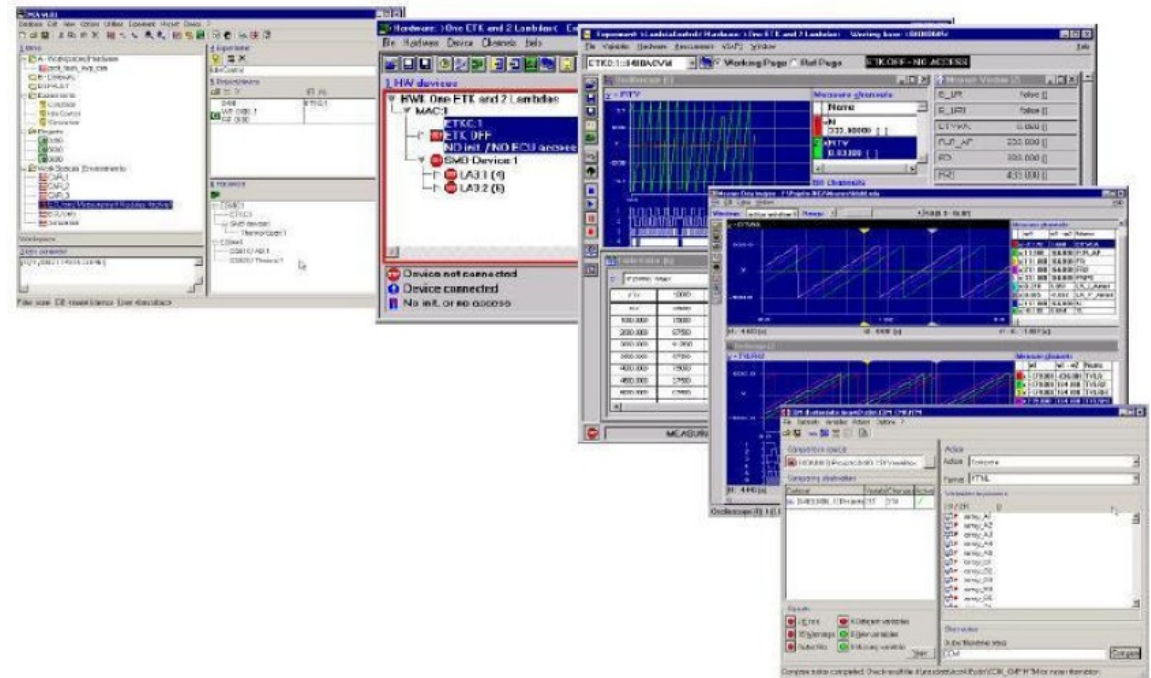




INCA - Structure of the tool

The main functionalities of INCA are:

- Calibration management
- Data acquisition
- ECU flash programming
- Measure analysis

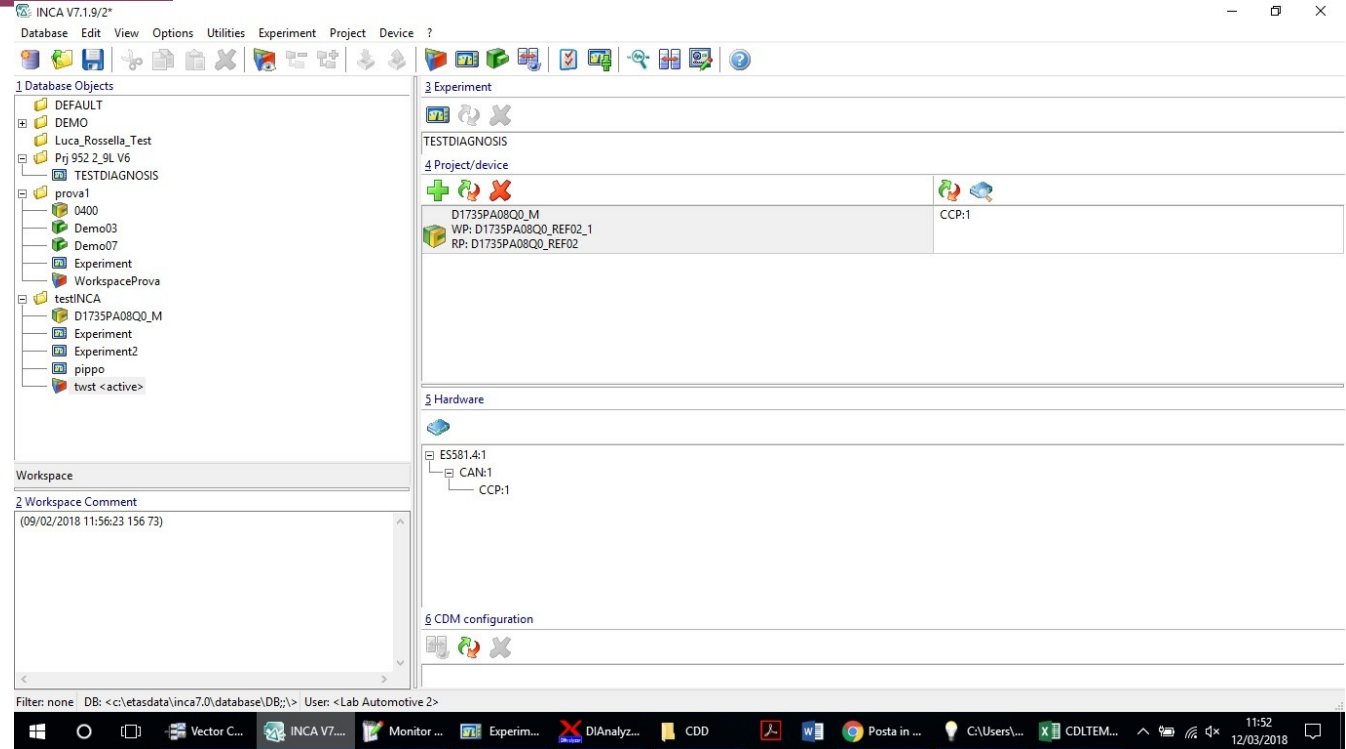




INCA - Configuration

The main screen of INCA has six macro areas:

- Database objects;
- Workspace comment;
- Experiment;
- Project/Device;
- Hardware;
- CDM configuration



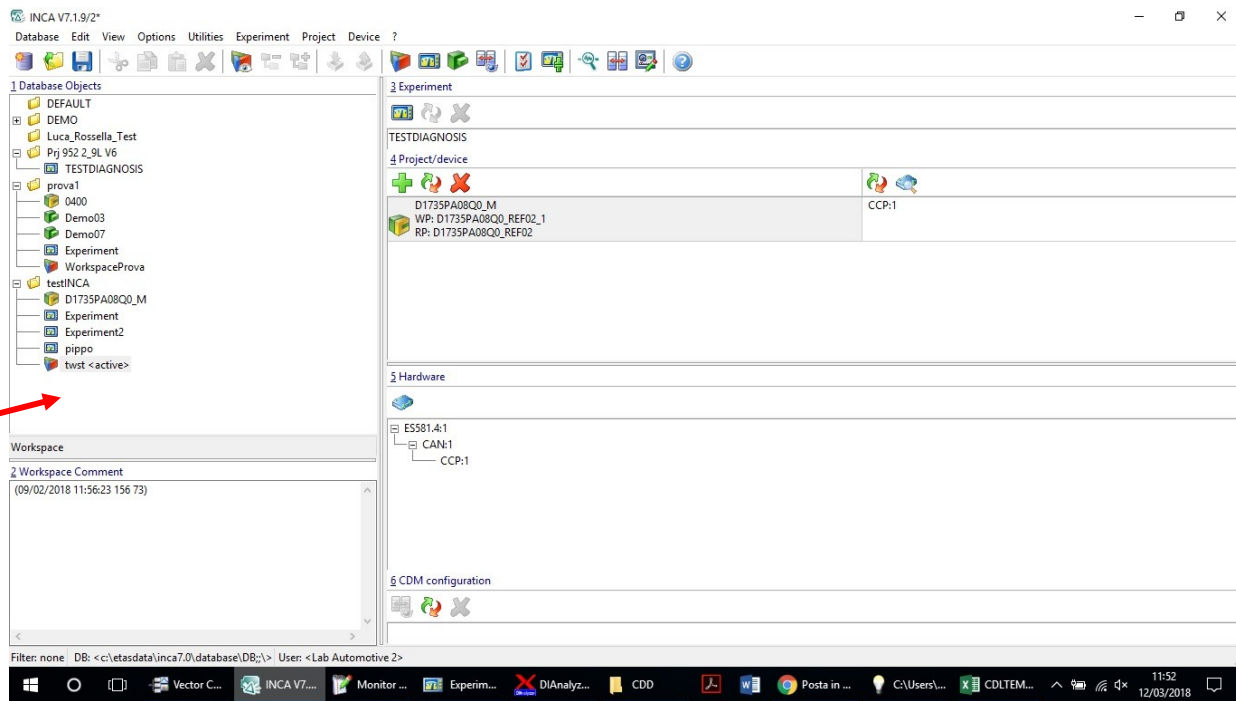


INCA - Configuration

The creation of a Top folder is the first step to connect INCA with any ECU.

Right button of the mouse on "Database Objects", and select

Add -> Top Folder



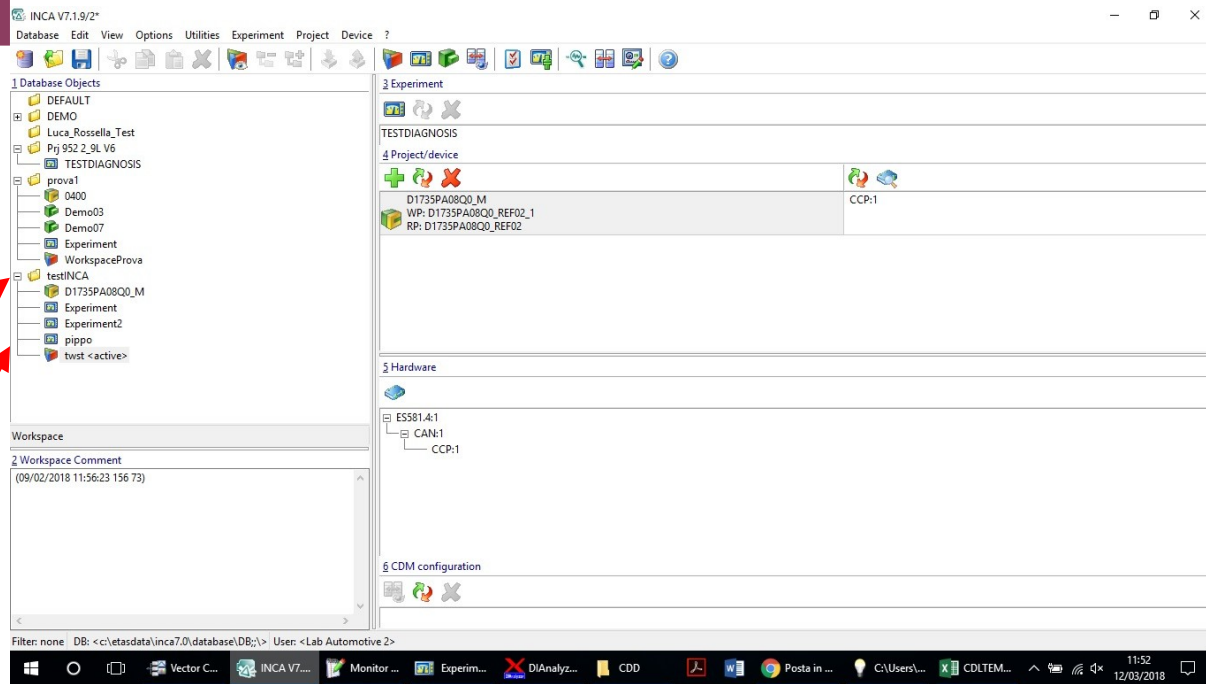


INCA - Configuration

After the creation of the Top Folder, the user has to create a Workspace. The workspace will contain all the information necessary to connect INCA with the ECU

Click on *Add Workspace* icon on the toolbar

Point the mouse on the created folder and, by clicking on the right button, choose *Add Workspace*





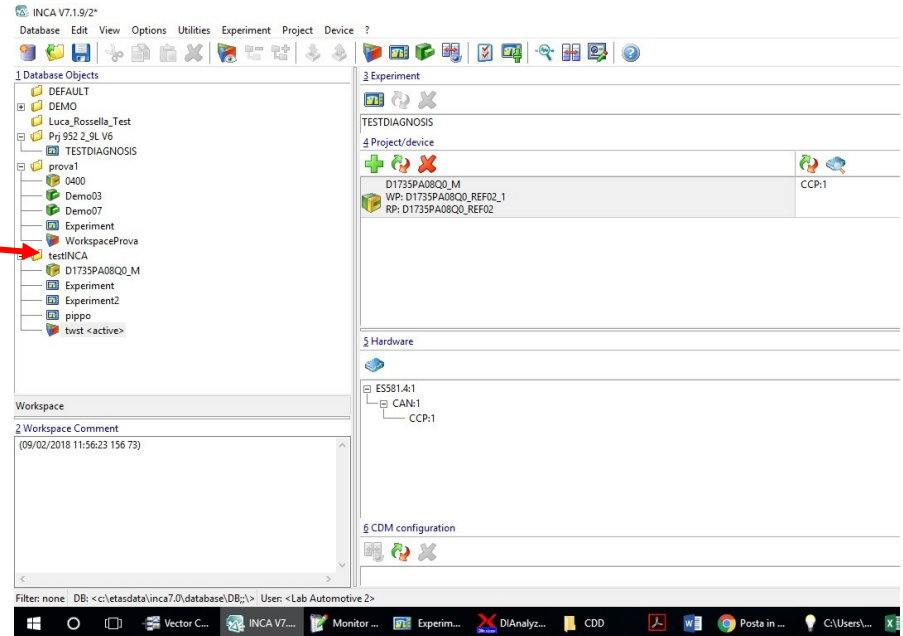
INCA - Configuration

INCA can communicate with an ECU only with a compatible software. In this way it's necessary the import of the ECU software in INCA. A software compatible with ASAP protocol is composed in two main parts:

- Base software: .A2L file;
- Dataset of the calibration: HEX, csv, s19, srec,...

Point the mouse on the created folder, click on the right button and choose **Add Ecu Project** and then follow the instructions to associate the A2L and HEX files

- **A2L**: file containing the detailed description of the elements of the sw, including physical characteristics of the parameters and the memory address where it is located
- **HEX, csv, s19, srec**: file containing the calibration dataset (the file format depends on the supplier)





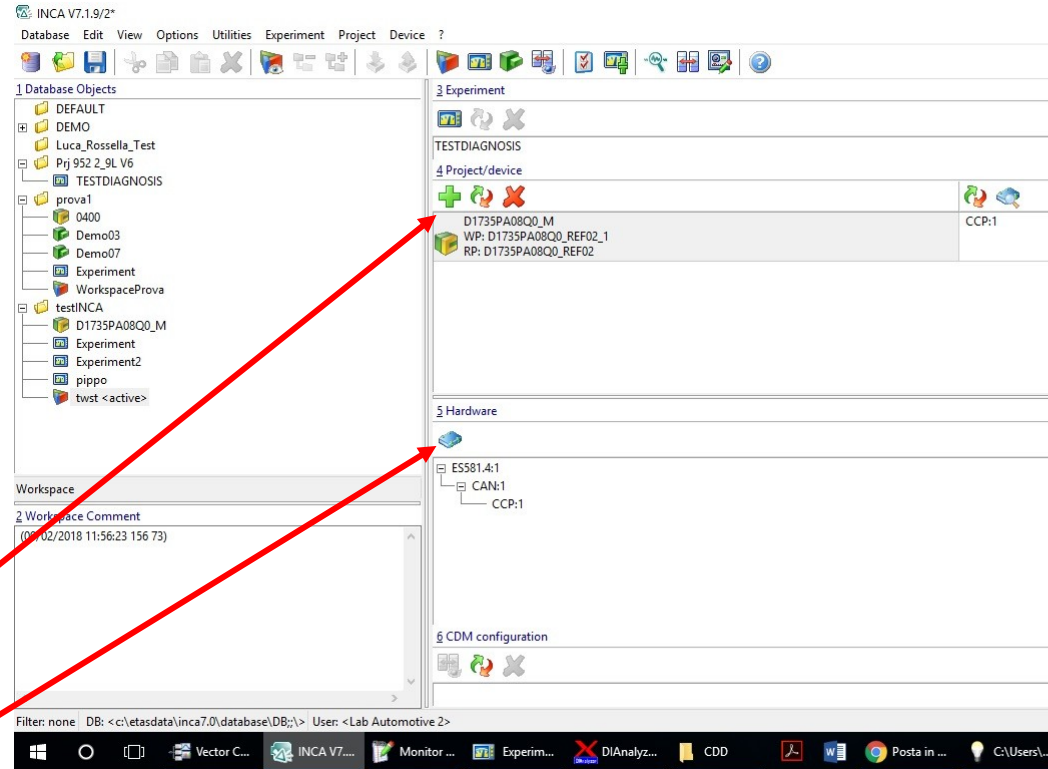
INCA - Configuration

INCA needs of a physical connection between host PC and ECU; user has to select the hardware device for communication; the selected hardware depends of the communication channel (ETK, CCP,...).

User has to associate (in the Workspace) the software and the hardware for the connection with the ECU

Select the workspace, in Project/Device section, press on “+” and add the software.

In hardware section, click with the mouse on Hardware Icon to select the device



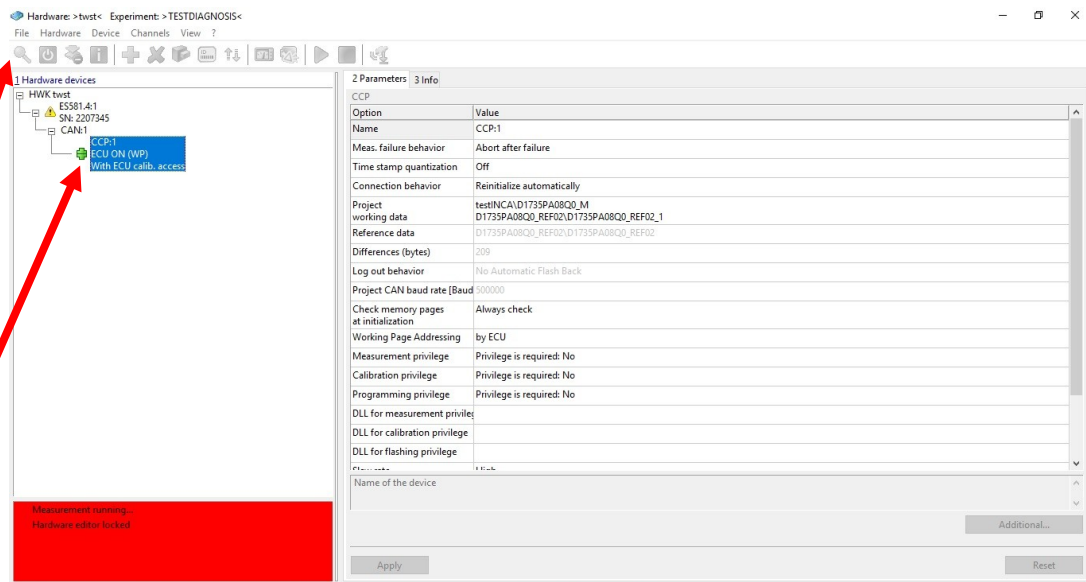


INCA - Configuration

Hardware section: in this section the user associates the connected hardware to the Project and so the workspace.

Clicking on the magnifier icon starts the search for the hardware interface

When the hardware has been identified and associated correctly, we will display the green icon shown in the figure (otherwise it will be red)





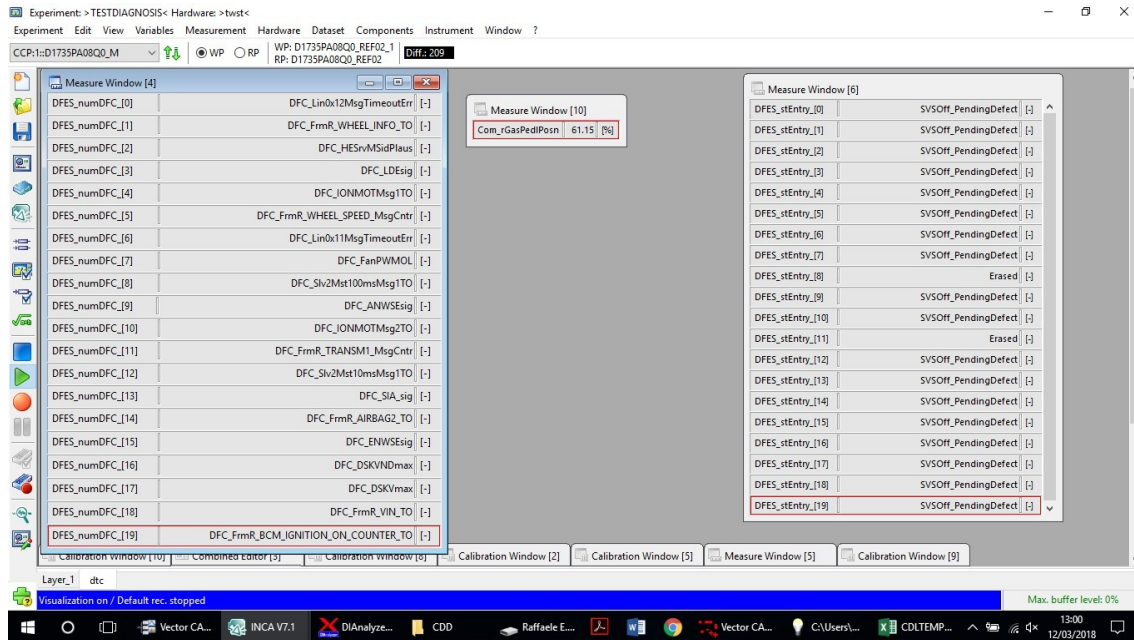
Working Page Vs Reference Page

INCA allows to pass in real time from the Working page to the Reference Page. The Reference Page is an only readable memory page; in the Reference Page there are the calibration provided from the software supplier.

The Working page is a writable memory page and the user can carry out the modification to the calibration; the modification of the calibration could be necessary for analysis procedure.

Reference page (RP): default calibration dataset contained in the sw supplied by the supplier

Working page (WP): dataset that can be modified during the development phase; useful to adapt the software for a specific application

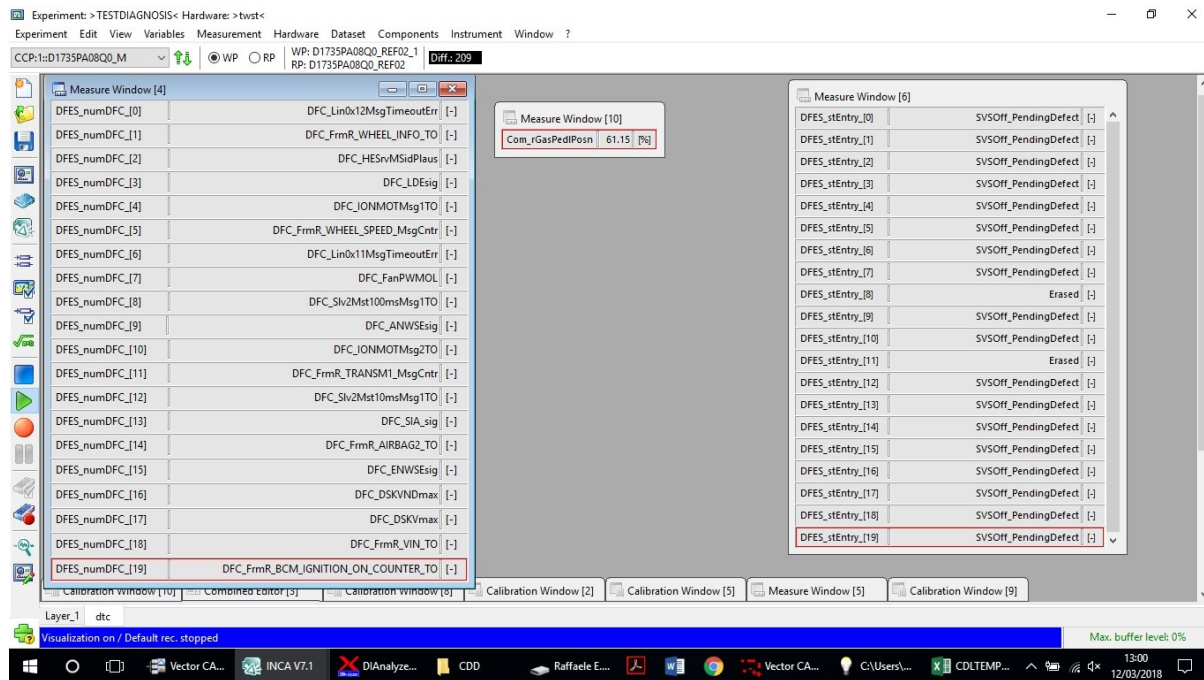




Calibration and Measurement

The calibration environment allows to read and modify software variables and parameters. This is the experiment section associated to the Workspace.

In the Experiment section, the user can check all the measurement and calibration variables.

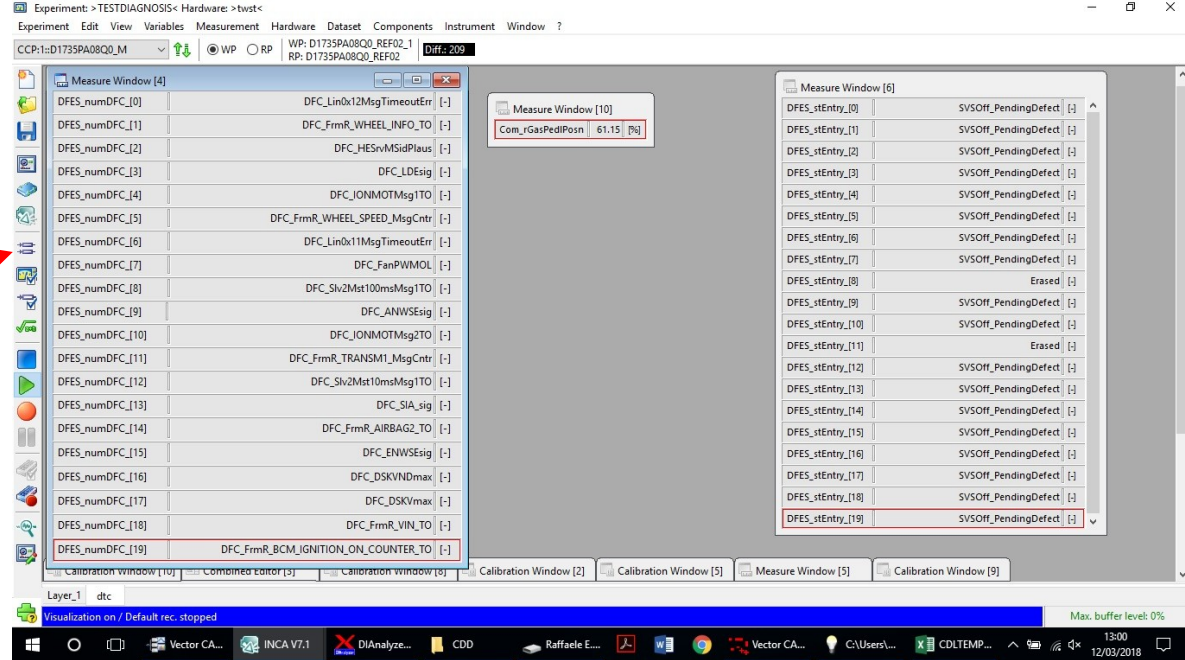




Experiment configuration

The user has to select the variables (measurement and calibration) that he needs for analysis and calibration tasks.

The user has to click on the *Variable selection* icon for variable selection.





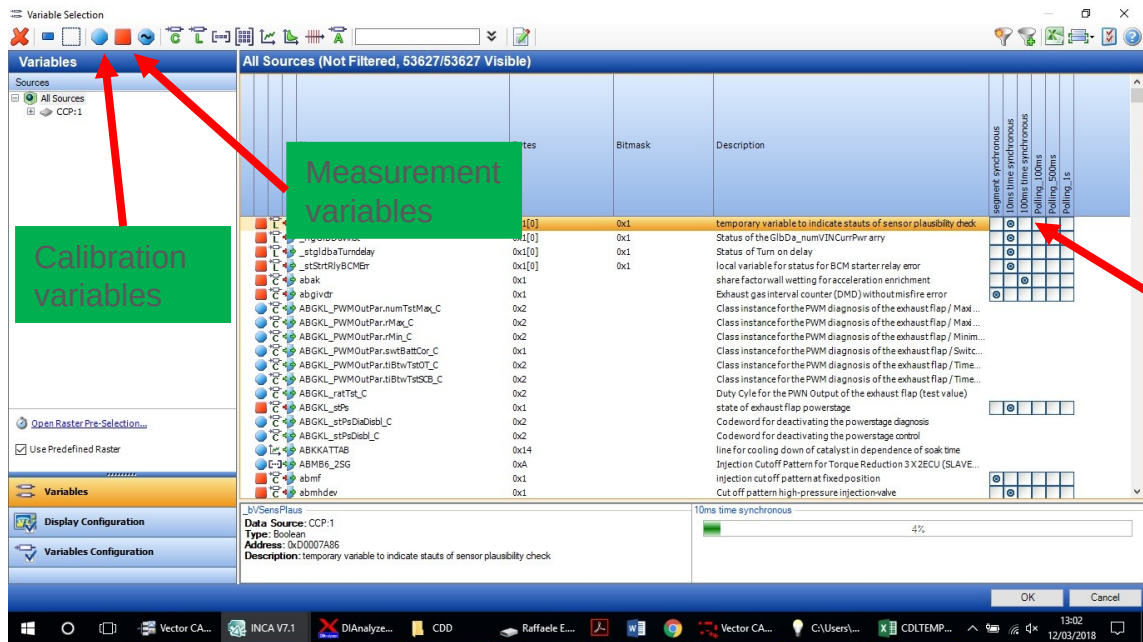
Experiment configuration

In the variable selection window there are all the calibration and measurement variable of the software.

Usually the supplier builds the software using many software modules; each variable is a part of a specific module.

The user can choose between measurement (e.g. raw data of a sensor or command value of an actuator) or calibration variables (e.g. Torque Map, enabling of a diagnosis,...).

For each measurement variable, the user can choose the acquisition raster.



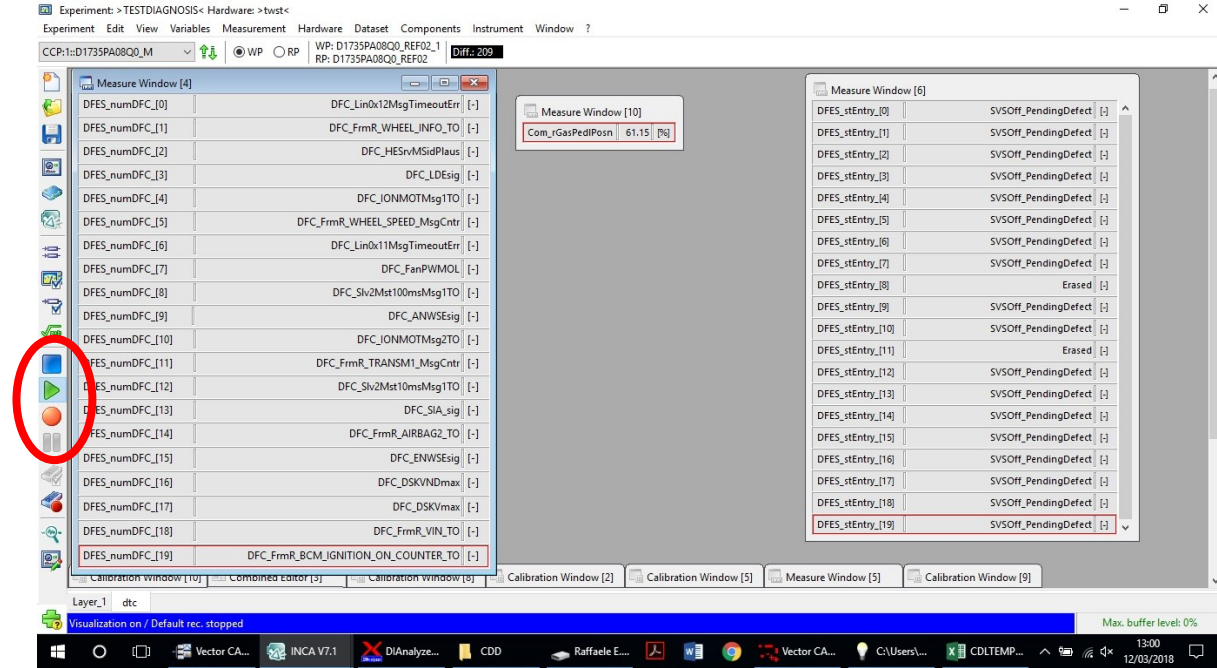


Experiment configuration

The user has to start the communication to check in real time the values of all selected software variables.

There are three commands:

- Start visualization – the user checks in real time the trend of software variables;
- Start recording – the user checks and records in real time the software variables;
- Stop measuring – the user stops the visualization and the recording.



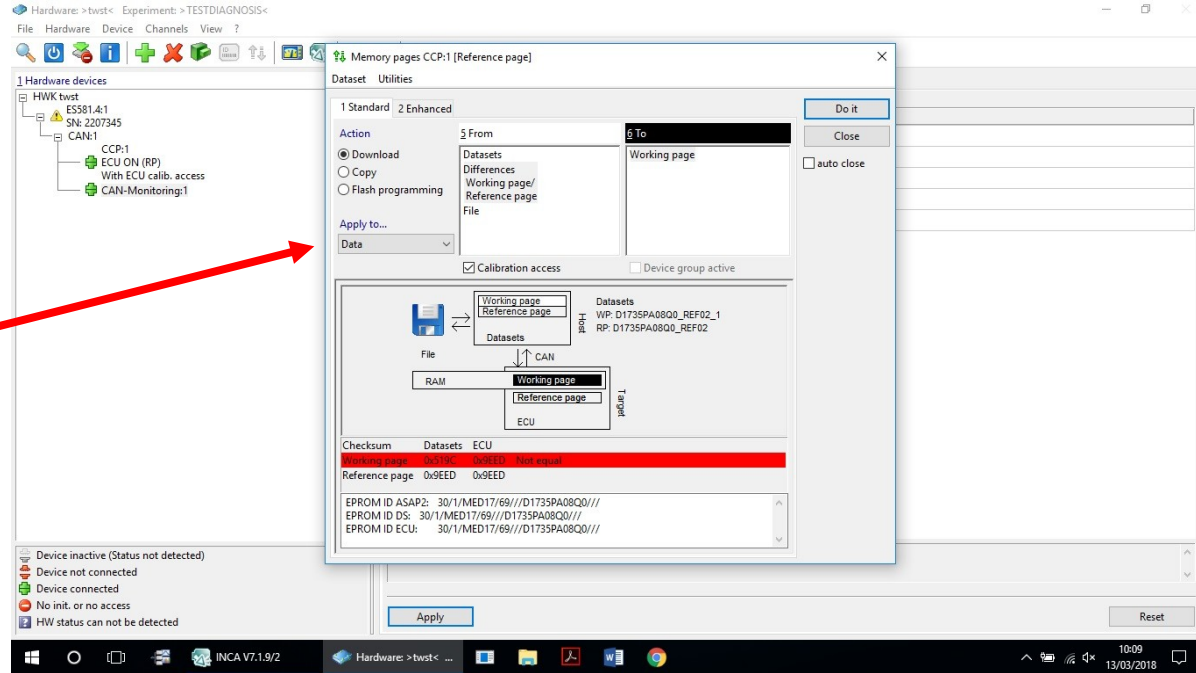


ECU flashing

INCA allows to perform the following operations on the control unit either by downloading the whole software or by flashing the variables dataset.

The Manage Memory section allows to perform the following operations:

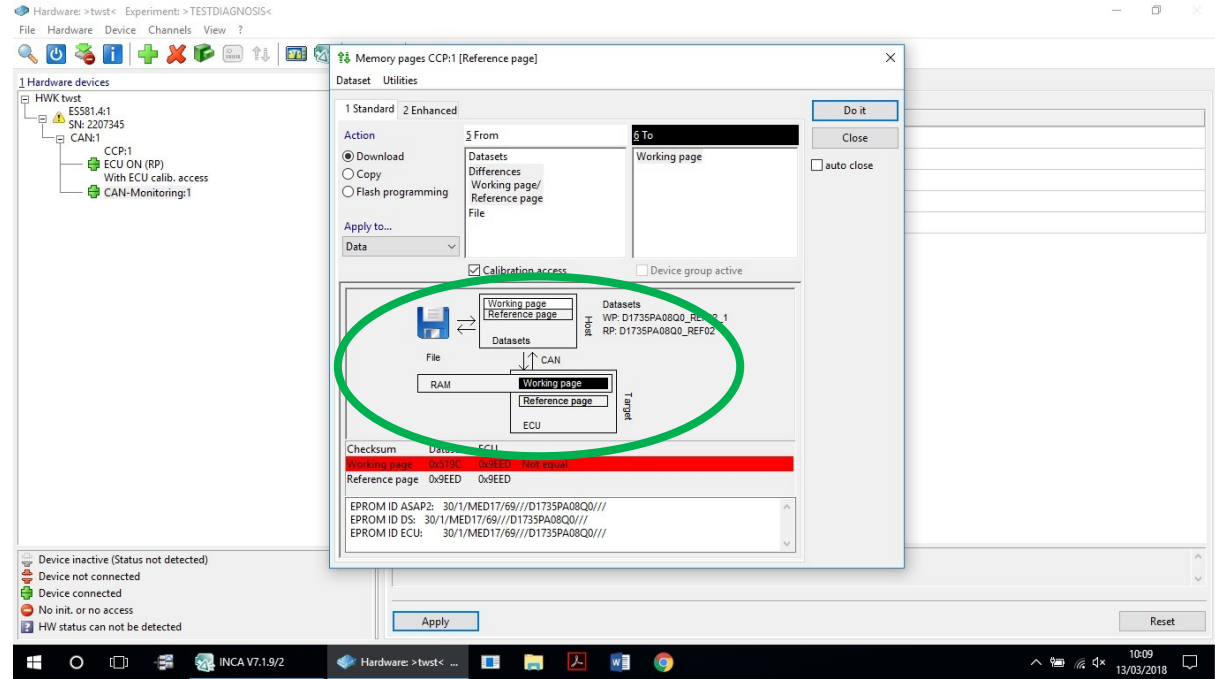
- Download
- Copy
- Flash programming
- Upload





ECU flashing

The Download of the ECU calibration data allows to download the data on the control page Ram





ECU flashing

Copy operation
the Working
Reference page

This operation is useful
in case of restoring
page.

Hardware: >twst< Experiment: >TESTDIAGNOSIS<

File Hardware Device Channels View ?

1 Hardware devices

- HWK twst
 - ES581.4-1
 - SN: 2207345
 - CAN:1
 - CCP:1
 - ECU ON (RP)
 - With ECU calib. access
 - CAN-Monitoring:1

Memory pages CCP:1 [Reference page]

Dataset Utilities

1 Standard 2 Enhanced

Action

- ☐ Download
- ☒ Copy
- ☐ Flash programming

Apply to...

Data

☒ Calibration access ☐ Device group active

5 From Reference page 6 To Working page

Working page Reference page

Datasets

File

RAM

Working page Reference page ECU

Checksum Datasets ECU

Working page	0x019C	0x0EED	Not equal
Reference page	0x9EED	0x9EED	

EPROM ID ASAP2: 30/1/MED17/69///D1735PA08Q0///
EPROM ID DS: 30/1/MED17/69///D1735PA08Q0///
EPROM ID ECU: 30/1/MED17/69///D1735PA08Q0///

Device inactive (Status not detected)
Device not connected
Device connected
No init. or no access
HW status can not be detected

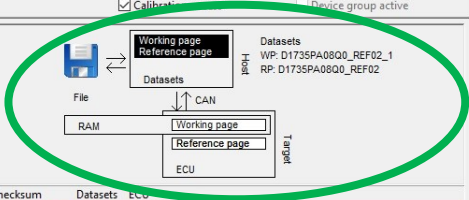
INCA V7.1.9/2

Hardware: >twst< ...

10:18 13/03/2018



The Upload operation is used to retrieve the Word documents from the Reference and store them in the memory of the client. The client can then use the CA database.

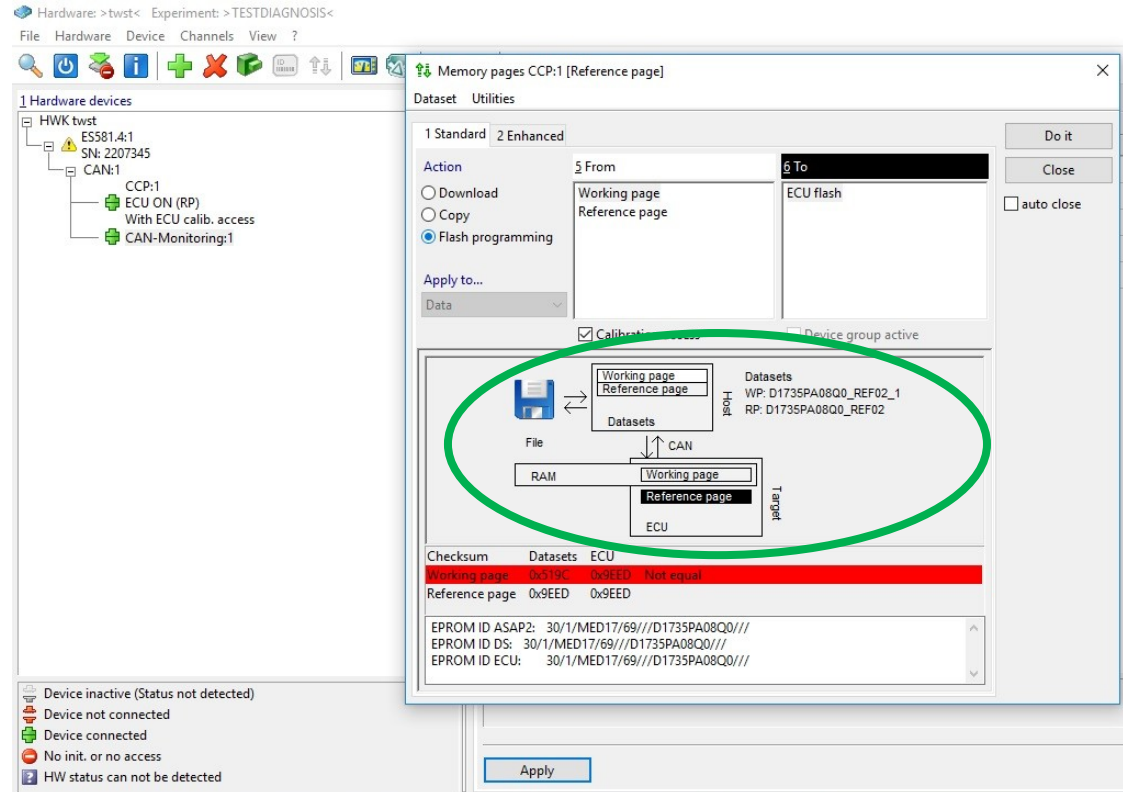




ECU flashing

The operation of Flash Programming requires to download the software and/or the calibration data to the permanent memory of the control unit.

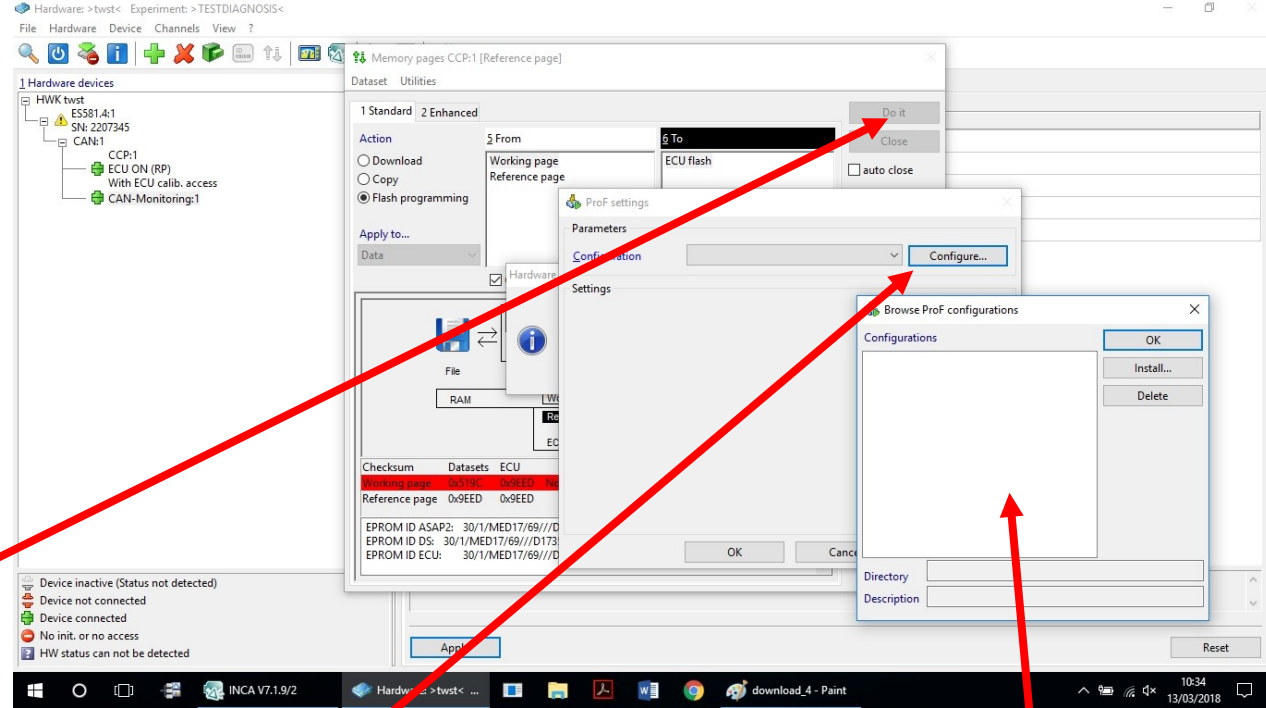
The operation of Flash Programming requires the configuration of the **ProF** which represents the configuration file of the control unit





ECU flashing

The user has to step reported in for ECU flashing.



1. Select **Flash programming** and press **Do it**

2. INCA open the ProF setting window; click on the button **Configure**

3. INCA open the ProF configuration window

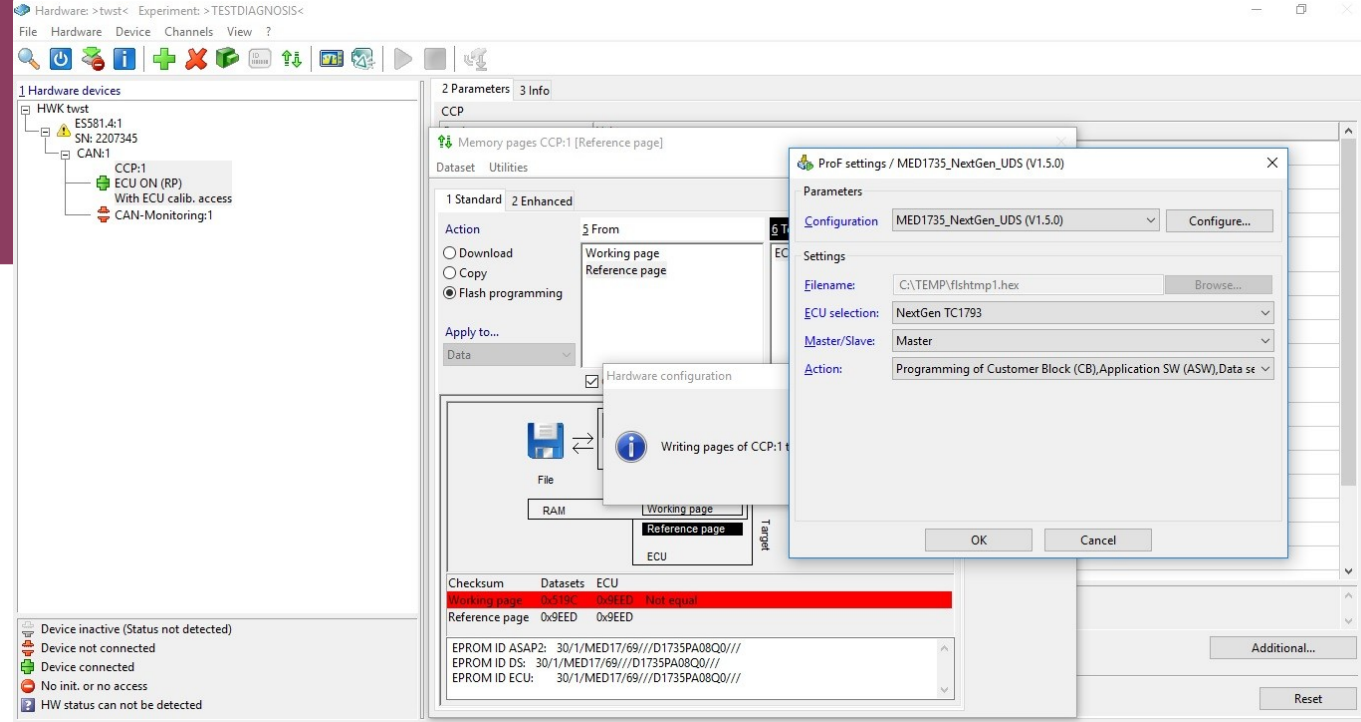


ECU flashing

After the selection of the ECU, the last step is starting the flashing procedure; the user must press the button

INCA starts to download the new software to the ECU.

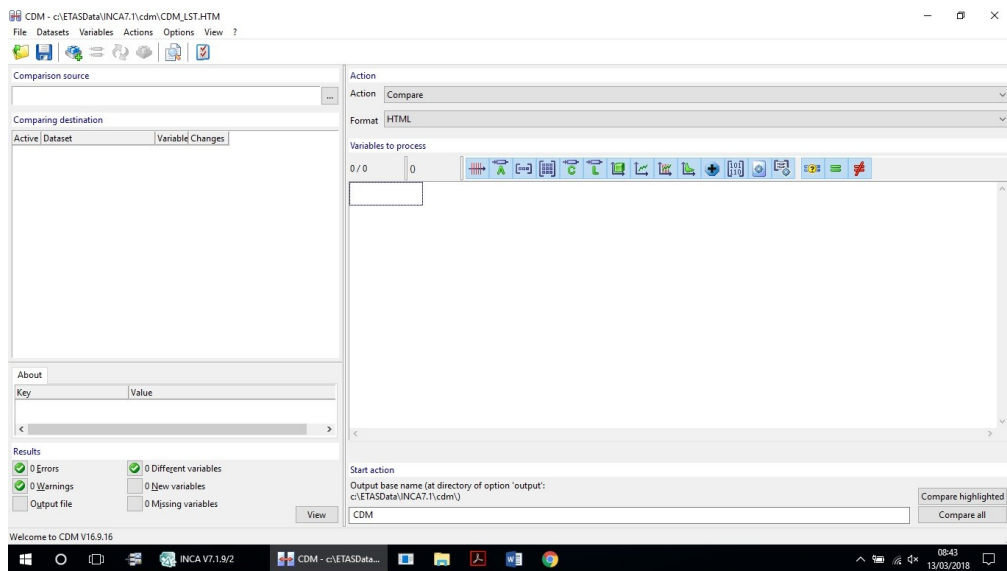
INCA will advise when the procedure finishes.



Calibration Data Manager

CDM

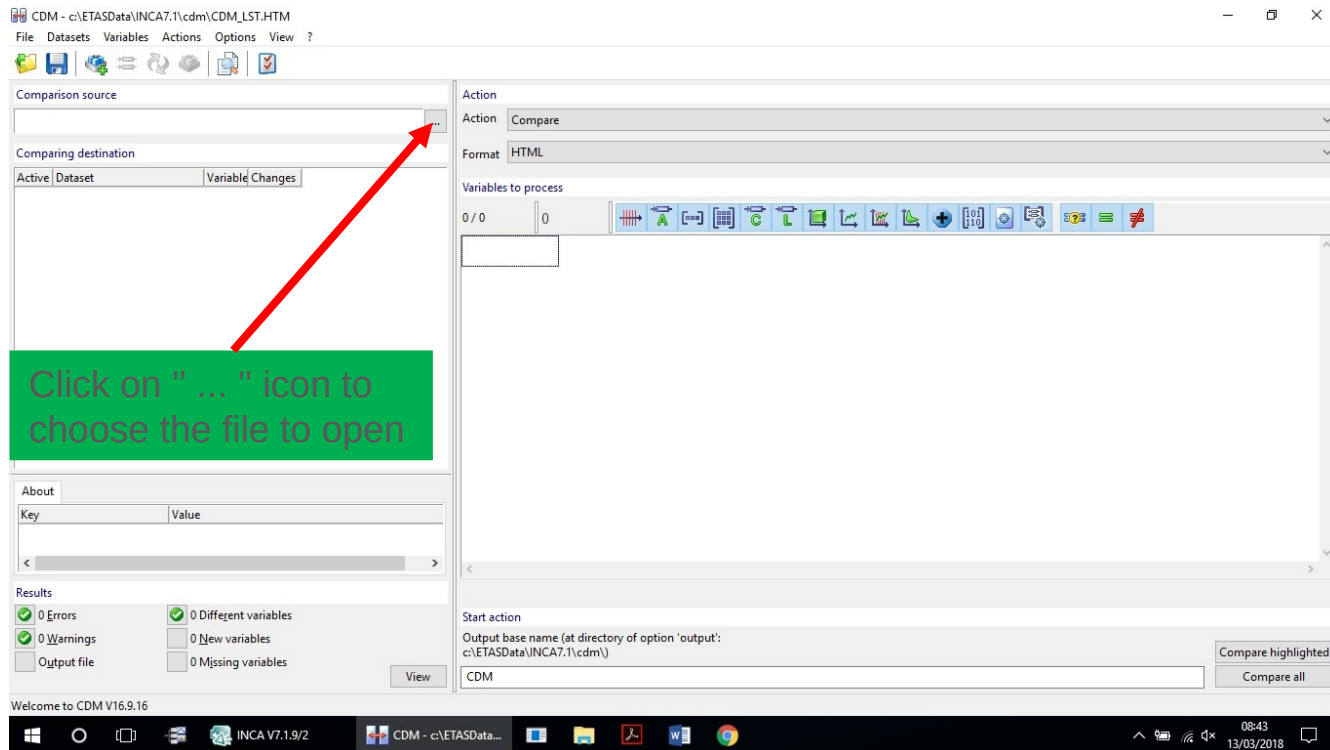
CDM is used to process calibration data sets in INCA. In particular, it allows to manage several datasets simultaneously through the Copy, List, Compare and Merge functions; it's possible also an export of the calibration data in Excel or ASCII format.





CDM: Calibration Data Manager

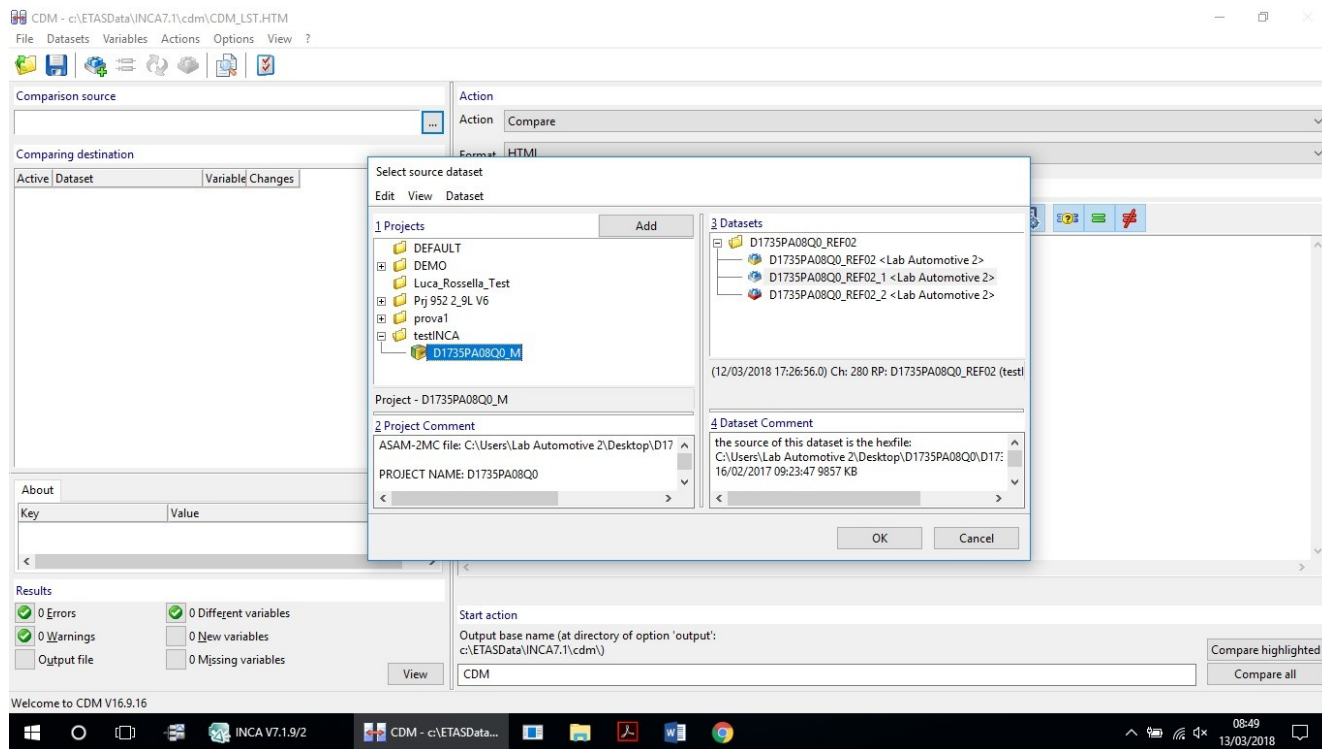
The user has to select the datasets that he needs to compare. The first step is the selection of the source; the source could be for example an .HEX file.





CDM: Calibration Data Manager

The user chooses the dataset that is in the INCA database.

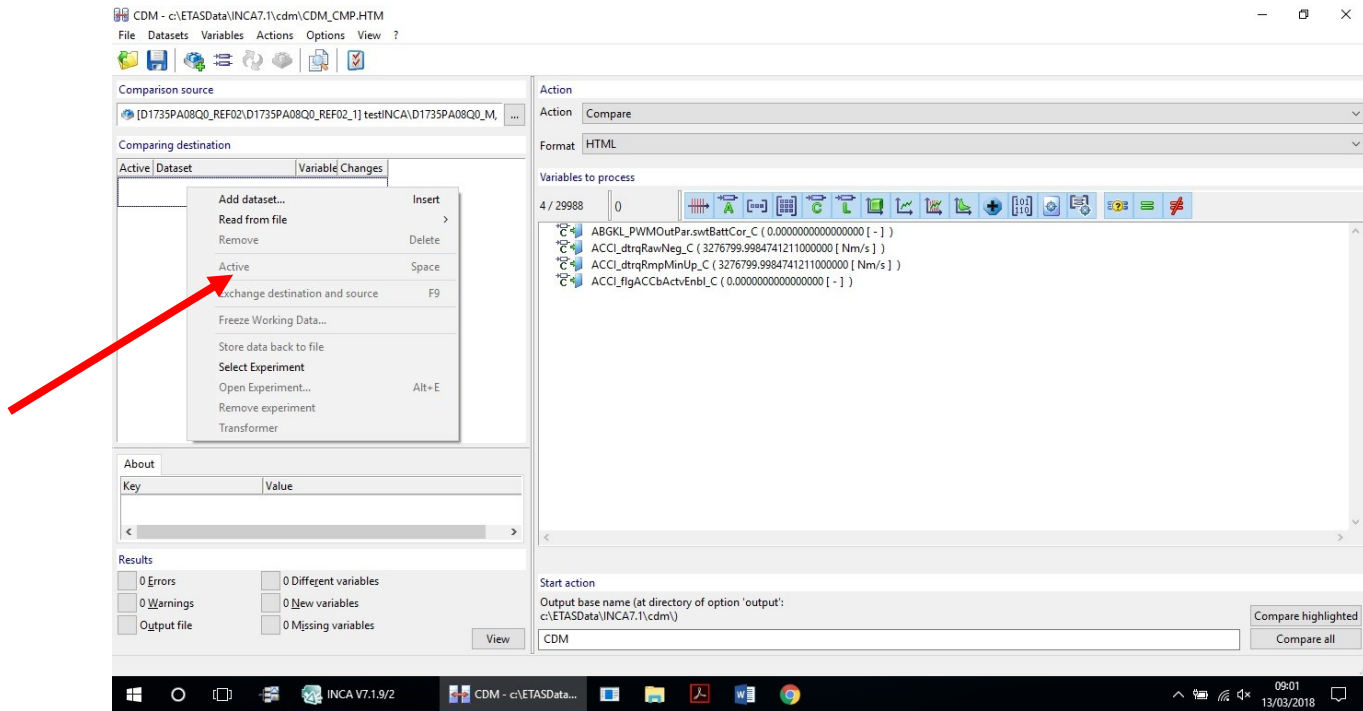




CDM: Calibration Data Manager

The user has to choose also the comparing dataset; the user has to select the data from database.

Click with the right mouse button in the window *Comparing Destination* and select *Add dataset*. Once you have chosen the dataset, click on it with the mouse to activate it

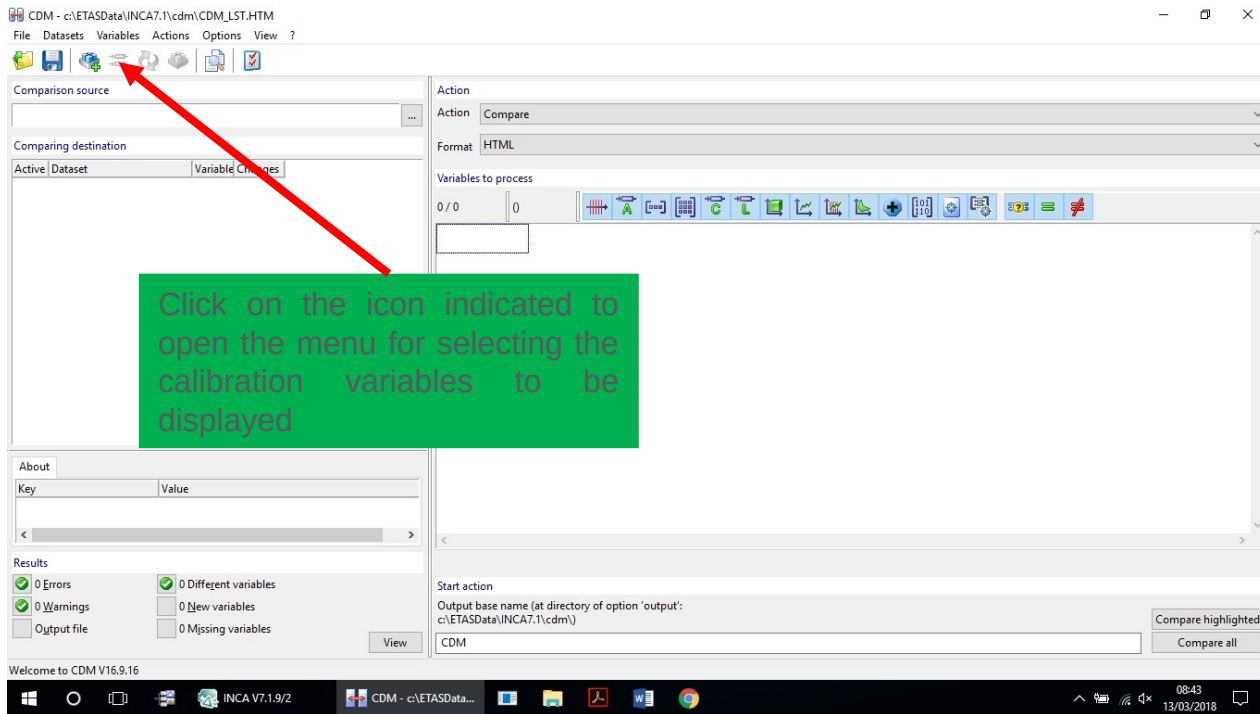




CDM: Calibration Data Manager

CDM can compare only a set of variable or the entire dataset.

In the first case, the user has to select the variables that he want compare.



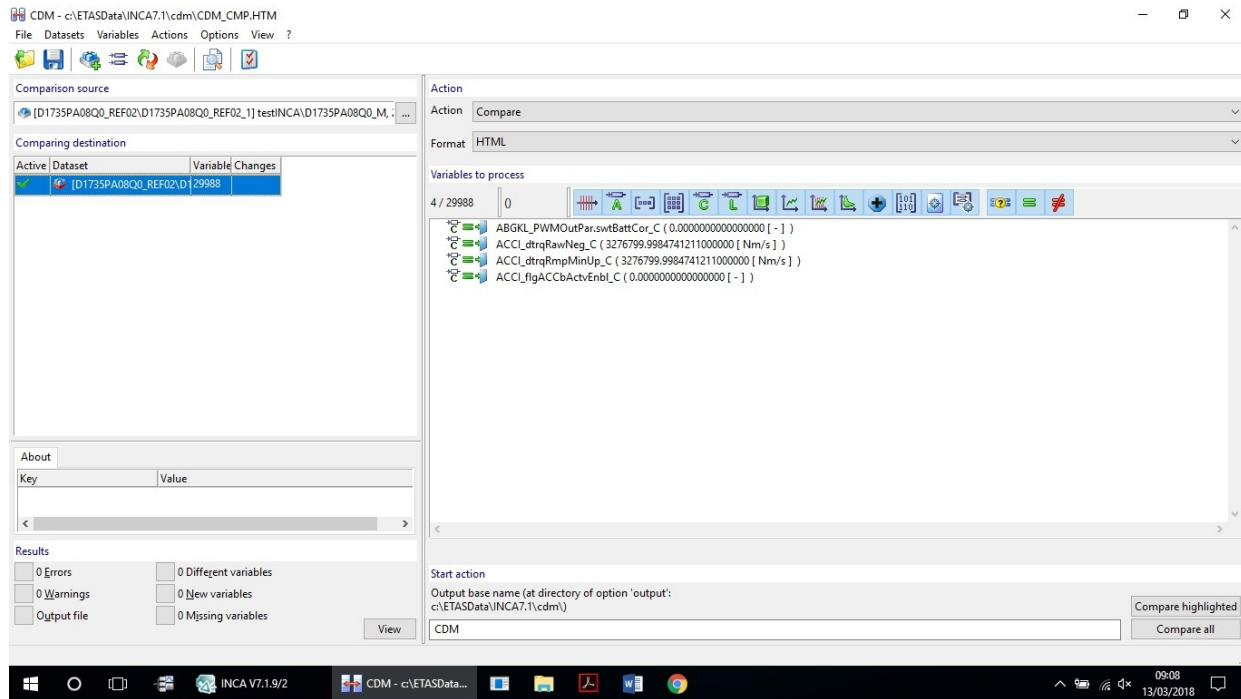


CDM: Calibration Data Manager

To compare the variables between the two datasets the user uses the button = or $\neq$.

The user can choose in Action windows the operation: Compare, Copy, List.

Furthermore it's possible to select the Format to save comparison data: HTML, ASCII,...

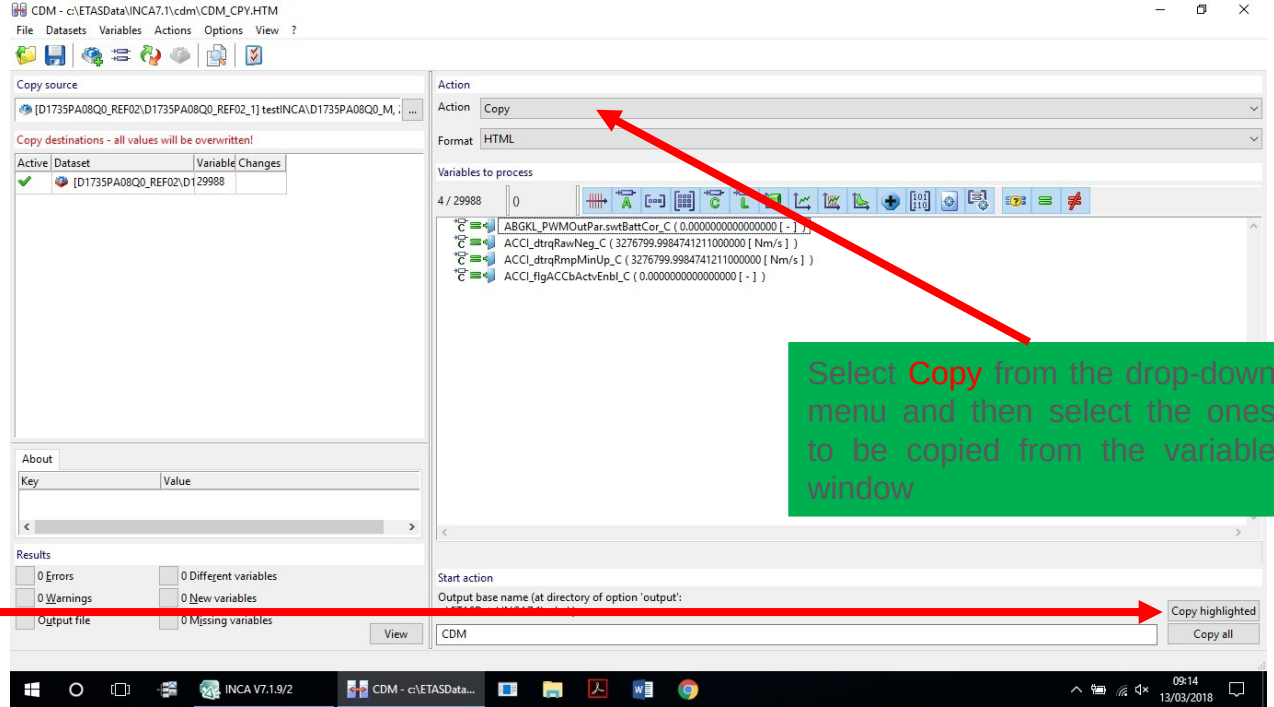




CDM: Calibration Data Manager

It is possible to copy variables between two datasets by selecting the appropriate function from the drop-down menu.

With these buttons the user can choose whether to select all the variables of the dataset or only those highlighted



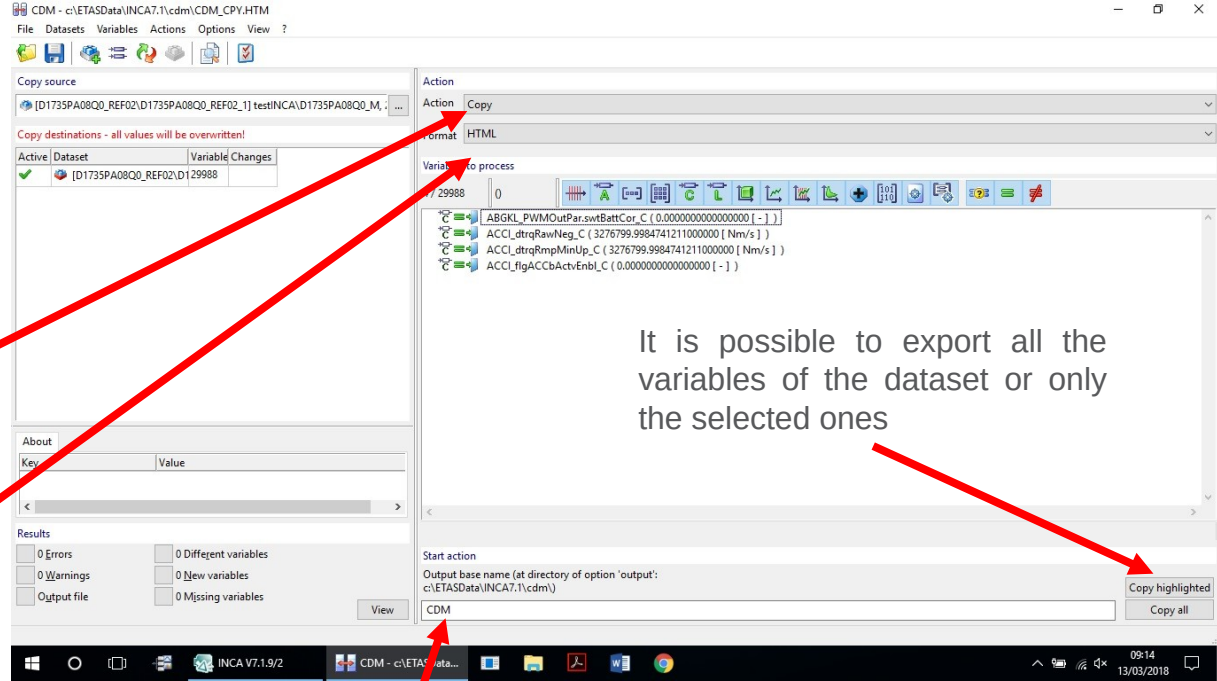


CDM: Calibration Data Manager

To export the dataset variables in Excel format (or HTML or ASCII) select the appropriate item from the drop down menu

Select **Copy** from the drop-down menu

User chooses the export format



It is possible to export all the variables of the dataset or only the selected ones

User chooses the name and path of the destination file



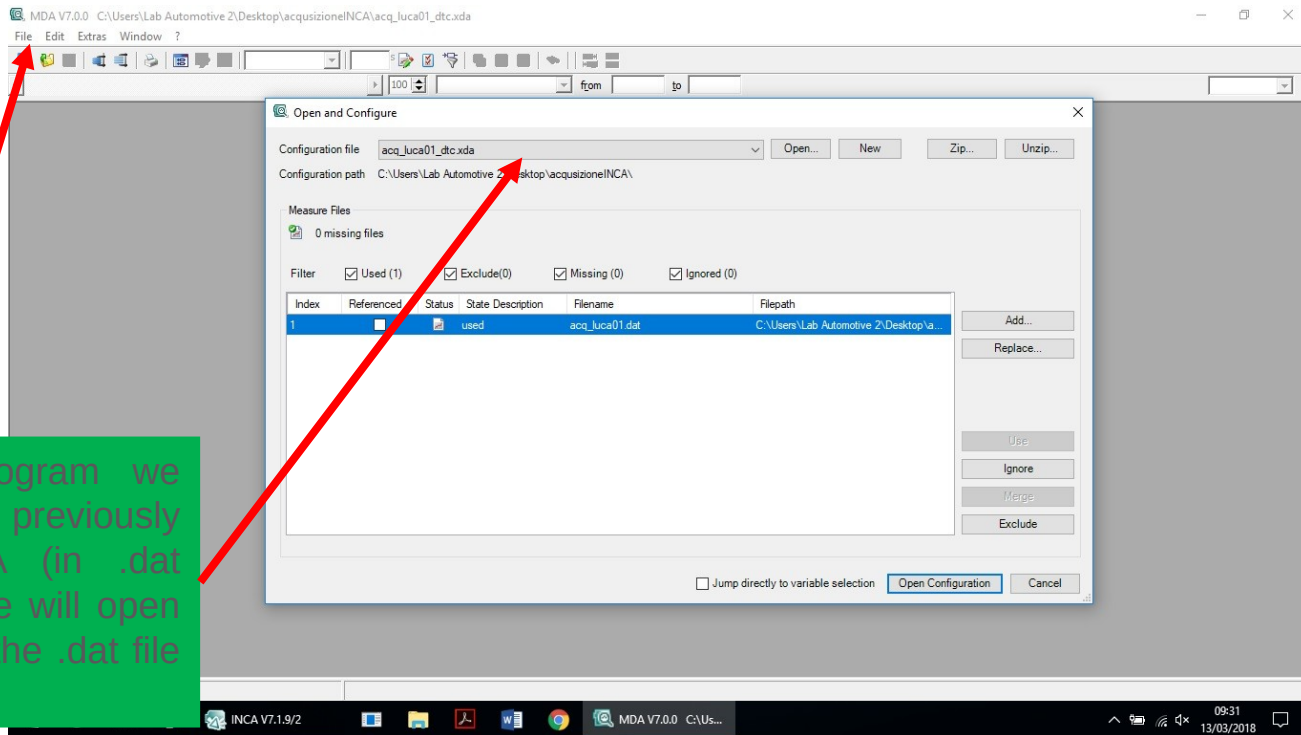
MDA



MDA: Measure Data Analyzer

MDA allows the post-processing analysis; the user can check only the variable that has been acquired in INCA.

Once open the program we choose the file previously acquired with INCA (in .dat format). Selecting File will open a window to choose the .dat file to use



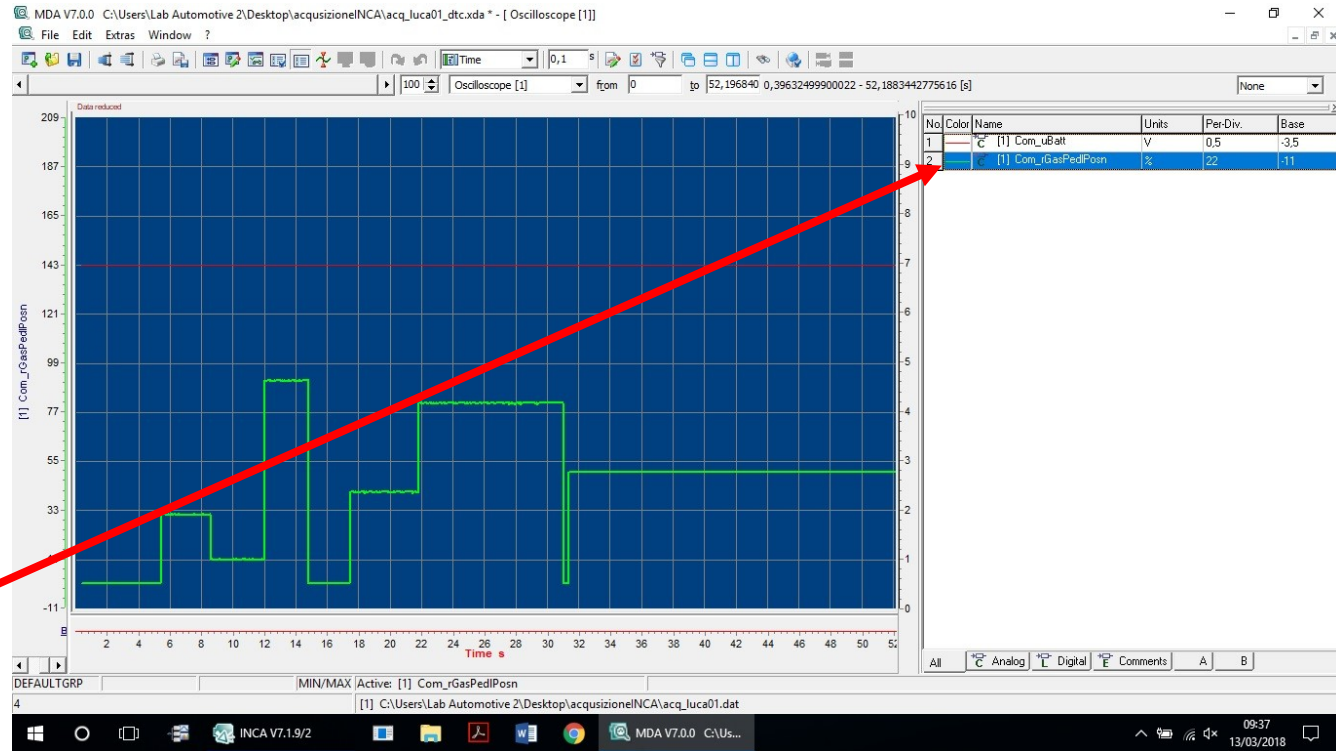


MDA: Measure Data Analyzer

Once the .dat file is open, the user chooses the variables to display from the window; the user chooses also the display mode.

The more comfortable way to perform the post processing analysis is using an oscilloscope: the user has to use the function *<new_oscilloscope>*.

To display the selected variables more clearly, the user can highlight them and can select the value *Distribute Uniformly*.



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